SASHIN DEEMANTHA

Software Engineering Intern

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PROFESSIONAL SUMMARY

Final-Year Software Engineering Undergraduate at SLTC Research University with practical industry experience spanning production web systems, enterprise ERP architecture, and AI-integrated applications. Proficient in modern full-stack development across the Python (FastAPI), TypeScript, React/Next.js, and Node.js ecosystems, supported by strong relational database design (PostgreSQL, MySQL) and automated CI/CD pipelines. Possesses verified experience bridging complex business logic with dependable system implementations from architecting state-machine audit trails in enterprise MRP software to deploying asynchronous AI platforms and live production web domains. Disciplined in Unix/Linux environments, automated testing, and clean, resilient code practices across the complete SDLC.

CORE TECHNICAL SKILLS

- **Languages & Core Foundations:** Python, TypeScript, JavaScript (ES6+), C# / Java (Academic Foundations), Go (Syntax & Foundations), HTML5, CSS3.
- **Frontend Engineering:** React.js, Next.js, Flutter (Cross-Platform Mobile), Tailwind CSS, Responsive Web Design, Component Architecture.
- **Backend & API Systems:** FastAPI, Node.js, Express.js, RESTful API Design & Integration, Asynchronous Architectures, JWT Authentication, Microservices.
- **Databases & Data Modeling:** PostgreSQL, MySQL, Relational Schema Normalization, Firebase (NoSQL), Data Integrity Constraints, Query Optimization.
- **DevOps, Cloud & Tooling:** Git / GitHub, CI/CD (GitHub Actions), Docker, Unix/Linux Environments, Postman, Jest (Unit/Integration Testing), SDLC & Agile/Scrum.
- **AI & Emerging Technologies:** Retrieval-Augmented Generation (RAG), LLM Orchestration, Prompt Engineering, AI-Assisted Developer Tooling (Claude Code, Copilot, Cursor).
- **Enterprise & Systems Design:** ERP/MRP Workflow Mapping, State-Machine Logic, Immutable Audit Trails, UI/UX Translation (Figma).

INDUSTRY EXPERIENCE

DI11 SOFT (PVT) LTD | UI/UX & Systems Architecture Intern

- ***June 2026 – November 2026***
- **Enterprise ERP Workflow Architecture:** Formulated system logic, data state transitions, and interaction flows for MicroMR, an enterprise Manufacturing Resource Planning (MRP) solution governing procurement, inventory, and sales lifecycles.
- **Event-Driven Audit Trails:** Designed and mapped immutable audit logs for Purchase Orders (POs) and Sales Orders (SOs), enforcing strict operational alignment between digital status changes and warehouse physical events (Goods Received Notes and Dispatch Notes).
- **Cross-Functional Collaboration:** Authored structured component dictionaries and functional requirement specifications to align complex backend relational constraints with intuitive, high-density frontend views.

FEATURED TECHNICAL PROJECTS

LankaLawBot | Web-Based Generative AI & Asynchronous Research Platform

- ***Lead Developer & AI Integrator*** | Python (FastAPI), React.js, Next.js, RAG & LLM Orchestration, PostgreSQL, REST APIs
- **AI & Agentic Pipeline:** Architected an end-to-end platform utilizing local Retrieval-Augmented Generation (RAG) and structured LLM workflows for semantic document analysis, transitioning beyond basic prompt wrappers into dependable software routines.
- **Asynchronous API Architecture:** Engineered low-latency asynchronous REST APIs using FastAPI connected to a responsive Next.js dashboard, efficiently managing streamed responses, payload validations, and connection timeouts.
- **Defensive Resilience:** Conducted rigorous edge-case testing across concurrent token generation and asynchronous retrieval states, eliminating client crashes and preserving system stability under load.

SAPSRI | Production Web Platform & Automated Deployment Pipeline

- ***Full-Stack & Database Developer*** | MySQL, JavaScript, PHP, Linux Environment, GitHub Actions (CI/CD)
- **Production SDLC Ownership:** Owned full-lifecycle development and maintenance for a live domain (sapsri.lk), authoring automated GitHub Actions CI/CD pipelines to achieve zero-downtime production deployments.
- **Relational Schema Engineering:** Normalized structured MySQL database schemas with strict referential integrity constraints, eliminating redundant anomalies across live organizational records.
- **Platform Hardening:** Diagnosed network fetching bottlenecks, mitigated web application security vulnerabilities, and configured remote production environments hosted on Unix/Linux servers.

EcoChange | Smart Cross-Platform System & Cloud Database

- ***Mobile App Developer & Full-Stack Contributor*** | Flutter, Firebase (NoSQL), Node.js Ecosystem, REST APIs, JWT
- **State Synchronization & Mobile Architecture:** Built a cross-platform mobile client in Flutter connected to cloud document storage, handling complex client-side state synchronization, geolocation features, and transaction ledgers.
- **Defensive Verification:** Configured token-based JWT authentication alongside boundary input validation routines to ensure data integrity and reliable API communication.

Telco Customer Churn & Automated Retention Engine | Containerized API Service

- ***Backend & MLOps Developer*** | Python, FastAPI, Docker, OpenAI API, REST APIs
- **Microservice Architecture:** Engineered a containerized Python backend service using FastAPI to preprocess tabular datasets and serve predictive inferences with sub-second response times.
- **Containerization & Deployment:** Built reproducible Docker environments for portable execution across Unix/Linux environments, managing dependencies and reproducible system builds.
- **Workflow Automation:** Integrated automated reasoning pipelines via OpenAI APIs to dynamically generate personalized retention actions from predictive model scores.

EDUCATION

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

SLTC Research University, Padukka **2022 – Present (Final Year)**

Relevant Coursework: Web Application Development, Database Management Systems, Data Structures & Algorithms, Object-Oriented Programming (Java/Python/C#), Software Architecture, Software Quality Assurance.